

FINAL PROJECT REPORT

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

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Team ID	[SWTID-2026-9584]
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Table of Contents

1. Introduction

- 1.1 Project Overview
- 1.2 Purpose

2. Ideation Phase

- 2.1 Problem Statement
- 2.2 Empathy Map Canvas
- 2.3 Brainstorming

3. Requirement Analysis

- 3.1 Customer Journey Map
- 3.2 Solution Requirement
- 3.3 Data Flow Diagram
- 3.4 Technology Stack

4. Project Design

- 4.1 Problem – Solution Fit
- 4.2 Proposed Solution
- 4.3 Solution Architecture

5. Project Planning & Scheduling

- 5.1 Project Planning

6. Functional and Performance Testing

- 6.1 Performance Testing

7. Results

- 7.1 Output Screenshots

8. Advantages & Disadvantages

9. Conclusion

10. Future Scope

11. Appendix

1. Introduction

1.1 Project Overview

This project automates the procurement and configuration of standard laptops for employees using ServiceNow Flow Designer. When an employee's Standard Laptop Request is approved, the automation automatically creates and assigns a hardware configuration task to the Hardware team, tracks the request through to delivery, and keeps the requester informed at every stage — replacing a process that today depends on manual email/chat handoffs.

1.2 Purpose

IT procurement currently lacks an automated way to generate and assign hardware configuration tasks for standard laptop orders. This causes delays, manual overhead for the IT team, and inconsistent turnaround for employees — particularly noticeable during onboarding, when a late laptop directly delays a new hire's first productive day. The purpose of this project is to design and plan a ServiceNow Flow Designer solution that removes this manual handoff, so laptops are configured and delivered promptly with minimal manual intervention.

2. Ideation Phase

2.1 Problem Statement

Using the “I am / I'm trying to / but / because / which makes me feel” framework, three customer problem statements were defined for the personas affected by the current manual process:

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1	a new employee joining the company	get my laptop configured and ready so I can start work immediately	I often wait several days without any update	there's no automatic link between my approved request and a task for the Hardware team	frustrated and unsure who to follow up with
PS-2	a member of the IT Hardware support team	configure and dispatch every approved request as soon as possible	I usually only learn about an approved request days later	there's no automated task creation or assignment step	reactive and overloaded
PS-3	a member of the IT procurement team	move requests smoothly from approval to configuration	I have to manually track and follow up on every request	there's no workflow automation connecting the request to a task	stretched thin, worried requests will slip through

2.2 Empathy Map Canvas

To understand the requester's experience before the automation, an empathy map was built for the “New Employee Requesting a Standard Laptop” persona:

Empathy Map — “New Employee Requesting a Standard Laptop”

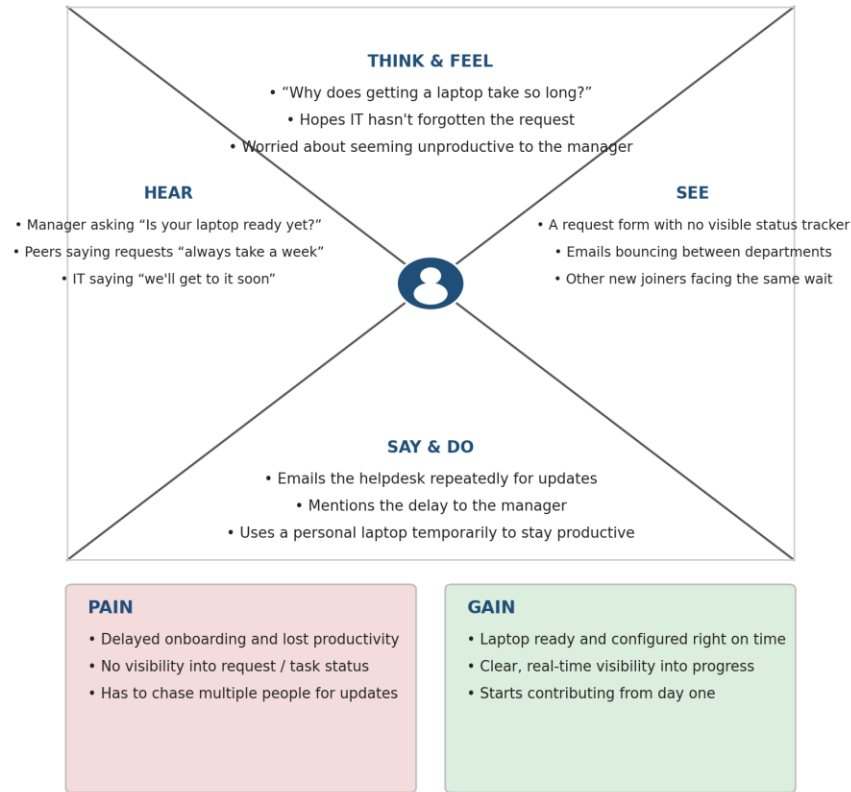


Figure 2.1: Empathy Map — New Employee Requesting a Standard Laptop

2.3 Brainstorming

The team brainstormed 15 ideas addressing the problem statement, grouped into four themes, then prioritized them by importance and feasibility:

- Cluster A — Flow Automation (Request → Task → Fulfilment): auto-create and auto-assign the hardware task, SLA escalation, CMDB auto-registration.
- Cluster B — Smart Approval Routing: cost-centre based routing, auto-approval under a threshold.
- Cluster C — Self-Service, Communication & Feedback: guided catalog form, Virtual Agent status checks, delivery notifications.
- Cluster D — Inventory & Vendor Readiness: buffer stock, vendor API integration, RPA purchase orders.

Idea Prioritization: Importance vs. Feasibility

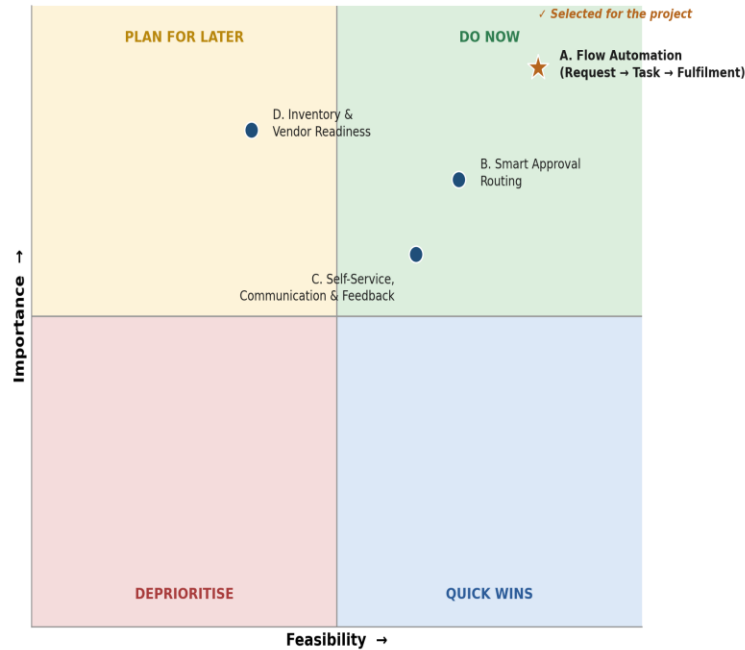


Figure 2.2: Idea Prioritization — Cluster A (Flow Automation) was selected for this project

Cluster A scored highest on both importance and feasibility, and directly matches the project's problem statement — it was selected as the idea to build via ServiceNow Flow Designer.

3. Requirement Analysis

3.1 Customer Journey Map

The current (as-is) journey for a standard laptop request was mapped across five phases — Entice, Enter, Engage, Exit, Extend — to surface friction points and map them to opportunities:

STREAMLINING IT PROCUREMENT

Customer Journey Map — Standard Laptop Request (Current / As-Is State)

SCENARIO: An employee (new hire or replacement case) needs a standard laptop, from the moment the need arises through approval, hardware configuration, delivery, and asset closure — before the Flow Designer automation described in this project.

	ENTICE		ENTER			ENGAGE		EXIT		EXTEND		
	How does someone become aware a laptop is needed?		How do they begin the process?			What happens in the core moments?		What happens in the process journey?		What happens after it's over?		
STEPS <i>Who does the person experience?</i>	New Hire Onboarding Trigger	Replacement Trigger	Browse Service Catalog	View Standard Laptop Item	Submit Request	Manager Approval	Manual Handoff to Hardware	Hardware Configures Laptop	Manual Status Update	Notification & Pickup	Asset Registration	No Structured Feedback Loop
INTERACTIONS <i>People, places, or things involved</i>	HR system, IT info, hiring manager	Manager, employee, internal chat / email	Employee Center portal, search bar	Catalog item page, generic request form	Catalog item, employee's own profile data	Manager, approval email / ServiceNow inbox	IT procurement staff, email chat to Hardware team	Hardware team, laptop, imaging tools	ServiceNow record, Hardware team member	Employee, informal message or in-person handover	CMDR, IT asset manager	None formal, occasional verbal comments
GOALS & MOTIVATIONS <i>"Help me..." / "Help me avoid..."</i>	Help me make sure I know a laptop is needed before day one	Help me get a working laptop without a long delay	Help me find the right item quickly	Help me understand what I'm requesting	Help me submit this with minimal effort	Help me review and decide quickly	Help me make sure Hardware finds out promptly	Help me know exactly what to configure and for whom	Help me reflect that the laptop is ready	Help me know the moment it's ready to collect	Help me keep accurate inventory records	Help me share how the experience went
POSITIVE MOMENTS <i>What works well today?</i>	Early heads-up gives IT some lead time	Manager is usually supportive of replacement	Catalog is easy to reach from the portal home page	A basic description of the laptop is available	The form itself is short and quick to fill	Manager can approve directly from email	A dedicated procurement person tracks handoffs	Hardware team is skilled and fast once they start	Status does eventually get updated	Employees are happy once they finally get the laptop	Assets do eventually get tracked	Employees do mention issues informally to their manager
NEGATIVE MOMENTS <i>What doesn't function today?</i>	No formal trigger — relies on someone remembering to email IT	No standard way to flag urgency or track the most data	Search returns multiple, on-foot laptop-related items	No visual guide comparing laptop tiers or specs	Department & manager fields must be typed in manually	Requester has no visibility into approval status	Delay if that person is busy, on leave, or forgets	Task details are often incomplete or scattered across emails	Update is a manual afterthought, often delayed	Employees often have to ask, or find out by chance	Registration is a separate manual step, sometimes skipped	No consistent data on satisfaction or turnaround time
AREAS OF OPPORTUNITY <i>How might Flow Designer improve this?</i>	Auto trigger a request from the HR onboarding record	A "Request Replacement" catalog option with reason capture	One clearly named "Standard Laptop Request" catalog item	Add a simple tier comparison (Basic / Developer / Power User)	Auto fill department & manager from the user's profile	Show real-time approval status in "My Requests"	Flow Designer auto-creates & assigns the task the moment it's approved	Auto-generated task carries all request details & an SLA	Closing the task auto-updates the request stage to "Delivered"	Automatic email / notification the moment the task closes	Auto-register the asset in the CMDB when the task closes	A short automated survey triggered after delivery

Figure 3.1: Customer Journey Map — Standard Laptop Request (Current / As-Is State)

3.2 Solution Requirement

Functional Requirements:

FR No.	Epic	Sub Requirements
FR-1	Laptop Request Submission	Submit via Service Catalog; auto-fill department & manager; select laptop tier.
FR-2	Approval Workflow	Route to manager; auto-approve under a cost threshold; reject with reason.
FR-3	Automated Task Creation & Assignment	Auto-create hardware task on approval; auto-assign by location; set priority/SLA.
FR-4	Status Tracking & Notifications	Auto-update RITM stage; notify requester by email; IT dashboard of status.
FR-5	Asset & Inventory Update	Auto-register the laptop in CMDB; update stock count after dispatch.

Non-Functional Requirements:

NFR No.	Requirement	Description
NFR-1	Usability	Simple guided form with auto-filled fields; clear status visibility.
NFR-2	Security	Role-based access; approval respects org hierarchy; every action logged.
NFR-3	Reliability	Flow triggers reliably on every approval, with retry on failure.
NFR-4	Performance	Task creation and notification complete within seconds of approval.
NFR-5	Availability	Inherits ServiceNow's 99.8% uptime SLA; no manual keep-alive step.
NFR-6	Scalability	Reusable Flow Designer subflows handle rising request volume.

3.3 Data Flow Diagram

The Level-1 DFD below traces how request data moves through the system, from submission to asset registration:

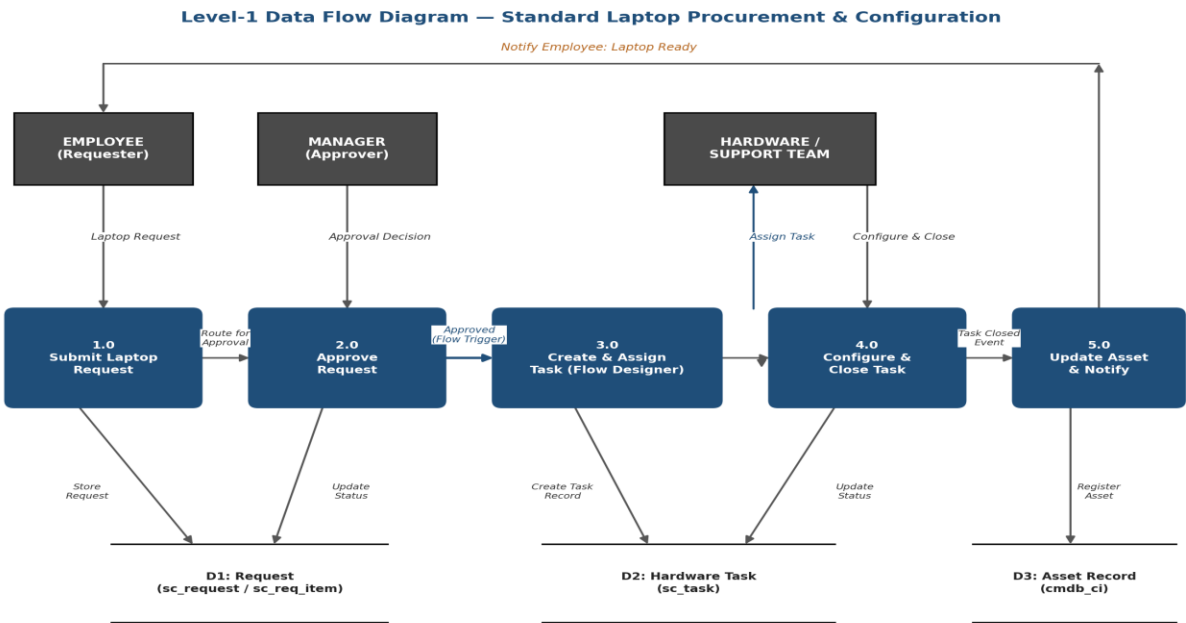


Figure 3.2: Level-1 Data Flow Diagram

3.4 Technology Stack

The solution is built entirely on native ServiceNow components, layered as follows:

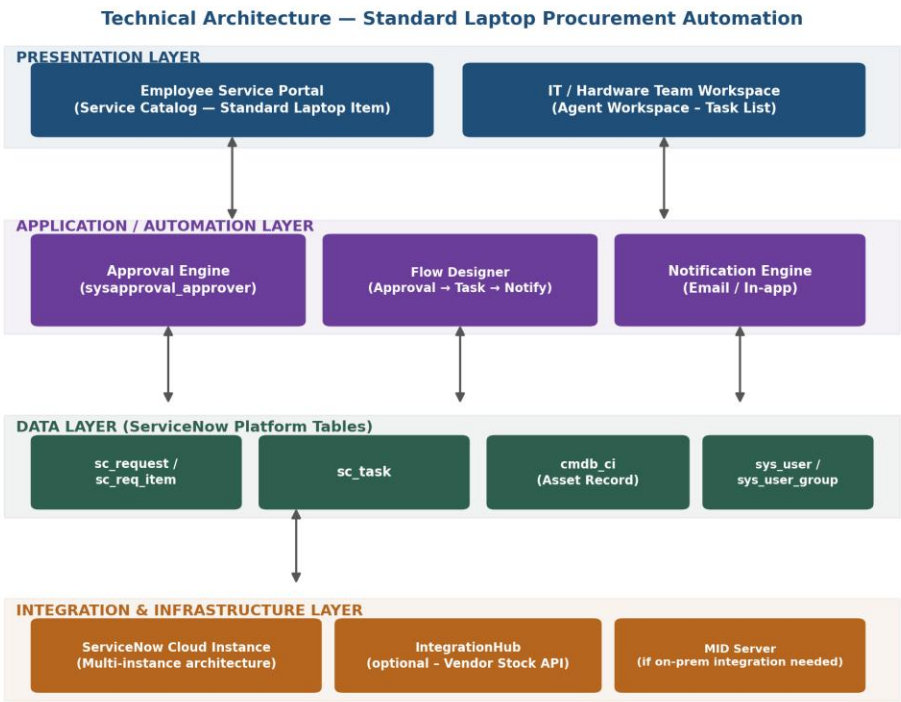


Figure 3.3: Layered Technical Architecture

4. Project Design

4.1 Problem – Solution Fit

The Problem-Solution Fit canvas confirmed genuine fit between the identified problem and the proposed automation:

- Customer Segment: Employees requesting a laptop, and the Hardware/Support team configuring it.
- Jobs-to-be-Done: Get a configured laptop to the employee quickly, without manual chasing.
- Root Cause: The catalog request has no automated link to a Hardware team task.
- Triggers: A new-hire record, a replacement request, or a stalled unassigned request.
- Your Solution: A Flow Designer flow that auto-creates and assigns the task on approval, updates status, and notifies the requester — removing the manual handoff entirely.

4.2 Proposed Solution

Problem Statement	IT procurement lacks an automated way to generate and assign hardware configuration tasks for standard laptop orders.
Idea / Solution	A Flow Designer automation that creates and assigns a hardware task on approval, tracks status, and notifies stakeholders.
Novelty	Uses native no-code Flow Designer — no custom scripting or external integration — with built-in SLA escalation and CMDB auto-registration.
Social Impact	Faster onboarding, less IT workload, higher employee satisfaction with IT services.

Business Model	Internal efficiency tool; value via cost-avoidance rather than revenue — fewer manual hours, faster turnaround.
Scalability	Reusable subflows extend the same pattern to other hardware/software requests and other departments.

4.3 Solution Architecture

The end-to-end component architecture and data flow are shown below:

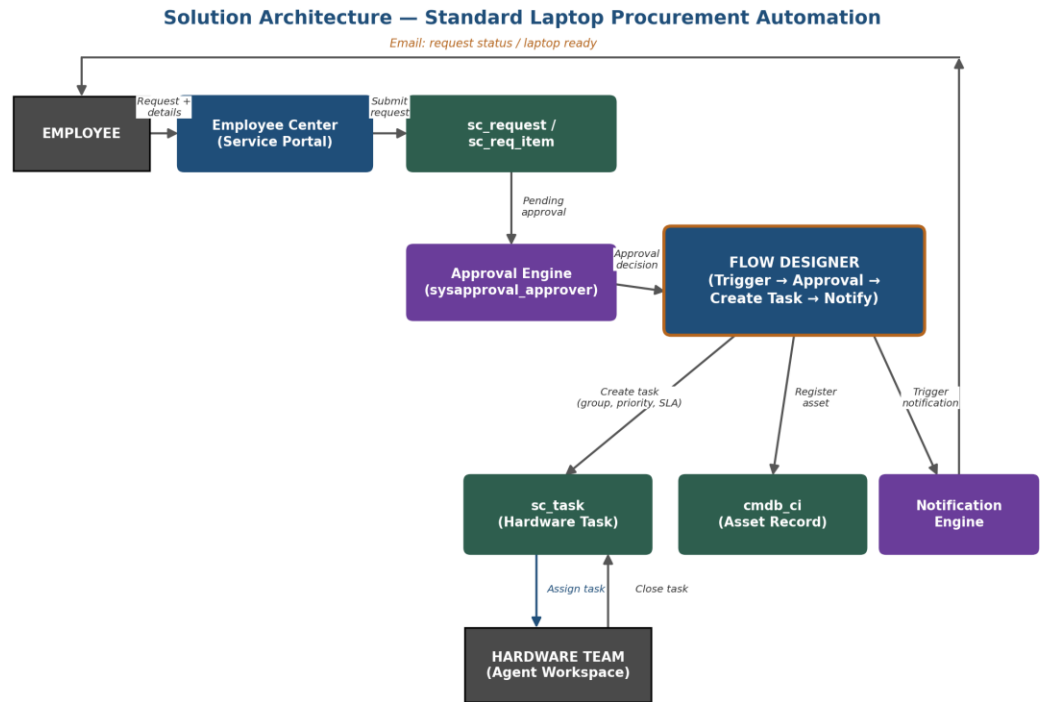


Figure 4.1: Solution Architecture — Standard Laptop Procurement Automation

5. Project Planning & Scheduling

5.1 Project Planning

The backlog of 10 user stories was estimated at 30 story points total and scheduled across two sprints:

Sprint	Focus (Epics)	Story Points	Duration	Planned Dates
Sprint-1	Laptop Request Submission, Approval Workflow, Automated Task Creation & Assignment	19	6 Days	28 Sep – 03 Oct 2026
Sprint-2	Status Tracking & Notifications, remaining Approval & Task items	11	6 Days	05 Oct – 10 Oct 2026

Average Velocity (Sprint-1) = 19 story points ÷ 6 days ≈ 3.2 points/day. The ideal burndown for Sprint-1:

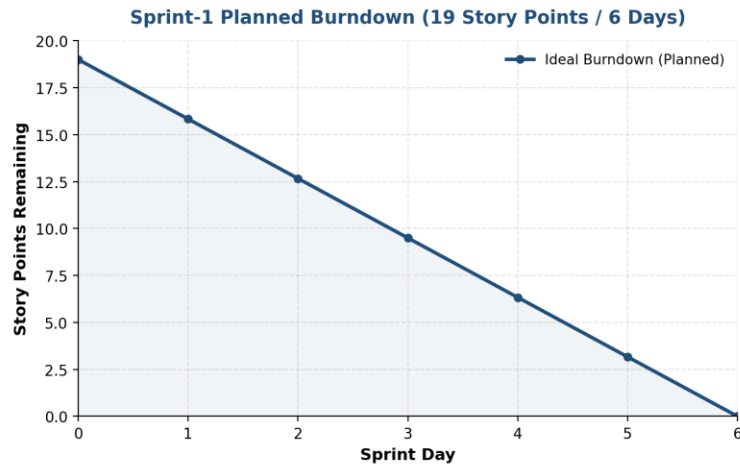


Figure 5.1: Sprint-1 Planned Burndown

6. Functional and Performance Testing

6.1 Performance Testing

User Acceptance Testing covers nine test cases spanning the full request lifecycle, including both positive and negative scenarios:

Test Case ID	Scenario	Type
TC-001	Submit a standard laptop request	Positive
TC-002	Manager approves the request	Positive
TC-003	Manager rejects the request	Negative
TC-004	Automated task creation & assignment	Positive
TC-005	Hardware team closes the task	Positive
TC-006	SLA escalation on delayed task	Positive
TC-007	Asset auto-registration in CMDB	Positive
TC-008	IT dashboard visibility	Positive
TC-009	Incomplete request submission	Negative

Full test steps, expected results, and the bug-tracking log are maintained in the separate User Acceptance Testing (UAT) document; results will be recorded there once execution begins.

7. Results

7.1 Output Screenshots

This project is at the design and planning stage; the Flow Designer flow, catalog item, and task automation described above have not yet been built in a live ServiceNow instance. Once implemented, this section will include screenshots of:

- The published “Standard Laptop Request” catalog item
- The Flow Designer canvas showing the trigger, approval, and task-creation actions

- A generated hardware task (sc_task) with auto-filled details
- The request record showing automatic stage updates

[Screenshots to be added after implementation]

8. Advantages & Disadvantages

Advantages:

- Removes the manual handoff between approval and hardware configuration
- Faster, more consistent laptop turnaround for employees
- Automatic SLA escalation reduces missed or forgotten tasks
- Built with no-code Flow Designer — quick to build and maintain
- Automatic CMDB registration improves asset tracking accuracy
- Pattern is reusable for other standard hardware/software requests

Disadvantages / Limitations:

- Requires an active ServiceNow license and platform access
- Only covers standard, pre-defined laptop configurations — exceptions still need manual handling
- Depends on accurate, up-to-date catalog and CMDB data to work correctly
- Requires re-configuration if the approval hierarchy or Hardware team structure changes
- Initial build and testing effort before benefits are realized

9. Conclusion

This project designs an automated, ServiceNow Flow Designer–based solution to the inefficiency identified during the ideation phase: standard laptop requests currently rely on a manual handoff between IT procurement and the Hardware team, causing delays and inconsistent turnaround. Through empathy mapping, problem-solution fit analysis, and requirement and architecture design, the project defines a solution that automatically creates and assigns hardware configuration tasks on approval, tracks status, and notifies stakeholders — directly addressing the original problem statement and user story. The sprint plan and UAT test cases provide a clear, ready-to-execute path from design to a working, testable automation.

10. Future Scope

- Integrate with a vendor API (via IntegrationHub) for real-time stock levels and delivery ETA
- Extend the same automation pattern to other standard requests — monitors, software licenses, mobile devices
- Add a Virtual Agent / chatbot for self-service status checks
- Explore ServiceNow Predictive Intelligence to auto-categorize or auto-prioritize incoming requests
- Roll out across additional departments and office locations

11. Appendix

Source Code (if any):

Not Applicable in the current phase — this solution is built using ServiceNow Flow Designer (no-code/low-code). Once implemented, the Flow and Catalog Item configuration will be exported as an Update Set. [Update Set link to be added]

Dataset Link:

Not Applicable — this project does not use an external dataset; it operates directly on live ServiceNow platform tables (sc_request, sc_task, cmdb_ci).

GitHub & Project Demo Link:

[To be added once the Flow Designer implementation and demo are complete]